



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
UNIVERSITY OF BARISAL

2nd Year 1st Semester Final Examination-2019, Session: 2018-19
Course Title: Object Oriented Programming

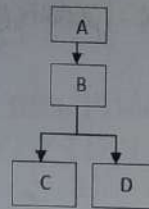
Course Code: CSE-2107

Time: 3 hours

Answer any five Questions from the followings. Parts of the same question should be answered consecutively. Marks: 60

1. (a) Why using procedural languages like C in building large scale Application are so uncomfortable and unmanageable? Why OOP languages such as Java have become so popular in this regard? Explain it from the perspective of three fundamental aspect of any OOP language. 4
- (b) In Object Oriented Programming abstraction and encapsulation both represent data hiding. Do you think they mean the same? Justify your answer with appropriate examples. 4
- (c) Write a function **power()** in Java to raise a number m to a power n . The function takes a **double** value for m and **int** value for n , and the returns the result correctly. Use a default value of 2 for n to make the function to calculate squares when this argument is omitted. Write a main function that gets the values of m and n from the user to test the function. 4
2. (a) What is static variable? Illustrate static method with proper example. What if main method is not **public static** in Java? 5
- (b) Why a class might provide a set method and a get method for an instance variable. Explain with an example. 4
- (c) What are different access specifiers in Java? Explain with example. 3
3. (a) Explain how the **Scanner** class can be used to take input from the user with an example. 3
- (b) Can we have more than one constructor in a class? If yes, explain the need for such a situation. 3
- (c) Write a class definition for the following scenario. You have to write a class called **MyFraction** which works on fractions of the form a/b where a represents the numerator and b represents the denominator. Both a and b are integers (i.e. if $a = 1$ and $b = 2$, then the fraction will be $1/2$). Your class should perform the following operations: 6
 - i. Create two constructors – (1) with no parameters that set $a = 0$ and $b = 1$; and (2) with 2 integer parameters that sets both a and b .
 - ii. Write an instance method for addition which returns the resultant object.
[int: $a/b + c/d = (ad + bc)/bd$]
 - iii. Write a static method for multiplication which returns the resultant object.
[Hint $a/b * c/d = ac/bd$]Write a static main() that allocates two Fractions, $1/2$ and $3/4$ and stores their sum in a third variable.
4. (a) What is Dangling-else problem? Determine the output for the given set of code when x is 9 and y is 11 and when x is 11 and y is 9. 3

```
if(x<10)
{
    if(y>10)
        System.out.println("*****");
    else
        System.out.println("#####");
    System.out.println("$$$$$");
}
```
- (b) Explain the difference between **break** and **continue** keyword. 3
- (c) Write a Java program that will perform the following operations: 3
 - a. Create an object of type ArrayList that will contain a list of floating-point numbers.
 - b. Now insert the following data: 12.34, 34.5, 5.6, 7.89, 10.12, 3.45
 - c. Show the number of elements in the object.
 - d. Remove 5.6 and 10.12Display the content of the object.
- (d) What do you know about java garbage collection and finalize method? Explain briefly. 3
5. (a) What do you know about inheritance in java? What are the advantages of inheritance? 4
- (b) Show how the following inheritance hierarchy can be implemented in Java. 3



- (c) Does Java support multiple inheritance? Justify your answer. 2
- (d) Explain the keyword **super** and function **super()** in Java with example. 3
- 6/ (a) What happens when a protected member is inherited as public? What happens when it is inherited as private? What happens when it is inherited as protected? 3
- (b) What is the difference between method overloading and method overriding? 3
- (c) Write a program to create a Database of the employees of an Organization with Name, Designation, Experience, Birth Date, Salary and Address information where Name, Designation and Experience are public members, Birth Date private member and other information are protected members. 6
7. (a) What is an Exception? Explain the syntax of try block and catch block with an example. 3
- (b) Explain the use of Finally block. 3
- (c) Consider a class named **BankAccount** that has three methods with preconditions: 6
1. The constructor (initial balance must not be negative)
 2. The withdraw method (withdrawal amount must not be less than the balance)
 3. The deposit method (deposit amount must not be negative)

Write code for the **BankAccount** class so that each of the three methods throws an **IllegalArgumentException** if the precondition is violated.

8. (a) What is a package? How can you define your own package? 3
- (b) What are the differences between abstract class and interface? 3
- (c) Consider the following code segment: 6

```

public interface Red {
    public void turnRed()
}
public interface Blue {
    public void turnBlue()
}

```

```

public class A implements Red {
    public void one(){
        System.out.println("A.1");
    }
    public void two(){
        one();
        System.out.println("A.2");
    }
    public void turnRed(){
        System.out.println("Nooooo!!");
    }
}

```

```

public class B extends A implements Red,
Blue {
    public void one(int x){
        System.out.println("B.1 "+x);
    }
    public void two(){
        System.out.println("B.2");
    }
    public void turnBlue(){
        System.out.println("Aaiiee!!");
    }
}

```

```

public class C extends A {
    public void one(){
        System.out.println("C.1");
    }
}

```

For each of the following, determine the output when the code will be executed. If some sort of errors are generated, give a brief explanation of the problem.

(a) `A a = new B();`
`a.two();`

(b) `B b = new B();`
`b.one(); b.one();`

(c) `Red r = new B();`
`r.turnRed();`

(d) `Blue b = new Blue();`
`b.turnBlue();`

(e) `C c = new A();`
`c.two();`

(f) `A a = new B();`
`a.turnBlue();`

(g) `A a = new C();`
`a.two();`